# Sagi **Shaier**

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## Research Interests

Artificial intelligence, knowledge graphs, commonsense reasoning, continual learning, dynamic systems, online learning, transfer learning

**Languages and Tools:** Python, PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, SQL, AWS, BigQuery

## Education

**University of Colorado Boulder** *Boulder, Colorado*

PH.D. IN COMPUTER SCIENCE *Aug. 2020 - Present*

**Kennesaw State University** *Kennesaw, Georgia*

BACHELOR'S OF SCIENCE *Aug. 2014 - Dec. 2018*

• Major in Computational and Applied Mathematics, Statistics minor, Pre-Med track. 3.96 GPA

## Professional Experience

**Quantum Metric**

Data Scientist - Research Team *Dec. 2019 - May. 2020*

- Spearheaded strategic projects such as predictive behavioral analytics, predictive churn risk score, loyalty score, & more
- Researched, constructed, and implemented various algorithms to predict customers' psychological nature across diverse industries such as airlines, retail, & banking
- Integrated Google's BigQuery, AutoML, & servers to optimized queries and analyses on big data while utilizing numerous supervised & unsupervised algorithms such as XGBoost, K-Means, Regressions, DBSCAN, PCA, Random Forest, & Neural Networks

**Kepler Computing**

Machine Learning Research Engineer (Contract) *July. 2019 - Oct. 2019*

- Identified key computation characteristics of AI workloads in areas such as self-driving, vision, speech, and AR/VR
- Worked with the chief computer architect and HPC engineers to co-design groundbreaking AI ASIC
- Studied cutting edge AI workloads and tracked innovations happening in AI

**Welocalize**

Machine Learning Intern *May. 2019 - Aug. 2019*

- Built regression models to predict project duration times and to detect outliers
- Gathered, engineered, and visualized various data types
- Identified bottlenecks within the organization using clustering analysis, Seaborn correlations, dimensionality reduction, and H2O.AI RF

**Hack Oregon**

Data Scientist (Disaster Team) *Feb. 2019 - Oct. 2019*

- Mapped the distribution of likely casualties in the 9.0 Cascadia Earthquake in order to better understand medical response needs

## Publications

2019 **Shaier**, Burke, “A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness)”

# Research Experience & Projects

### Human African Trypanosomiasis

*Kennesaw State University*

*Aug 2015 - Dec. 2018*

- Pioneered a nine-dimension disease propagation matrix and discovered two factors that offered strong evidence for the eradication of the disease

### Waterbender

Water drops detection and removal algorithm

*github/Waterbender*

*Sep. 2019*

- Developed a model to enhance the performance of image-based driving assistance systems in rainy weather

### DNet

Modified Yolo V3 model to detect objects and their distance from the camera

*Video*

*Jun. 2019*

- A real-time detection model utilizing only a single, simple camera

# Military

### Israel Defense Forces

*Israel*

*Aug 2010 - Aug. 2013*

- Prevented further escalations in terrorism throughout the Middle East

# Awards & Presentations

## Awards

Summer 2021 **Dean's Summer Research Fellowship** University of Colorado Boulder

Fall 2020 **Early Career Professional Development Fellowships** University of Colorado Boulder

Fall 2020 **Departmental Fellowship** University of Colorado Boulder

Fall 2017, Spring 2018, Fall 2018 **National Science Foundation Scholarship** Kennesaw State University

Summer 2018 **Society for Mathematical Biology Award** Kennesaw State University

Summer 2018 **Birla Carbon Scholarship** Kennesaw State University

Spring 2018 **Undergraduate Research Award** Kennesaw State University

## Presentations

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| Jul. 2018 | A Mathematical Model for the Effect of Domestic Animals on the Basic Reproduction Number of Human African Trypanosomiasis (Sleeping Sickness) | Annual Meeting of the Society for Mathematical Biology in Sydney, Australia |
| Apr. 2018 | A Mathematical Model for the Effect of Domestic Animals on the Basic Reproduction Number of Human African Trypanosomiasis (Sleeping Sickness) | KSU Symposium of Student Scholars |
| Jul. 2017 | A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness) | Annual Meeting of the Society for Mathematical Biology in Utah |
| Apr. 2017 | A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness) | KSU Symposium of Student Scholars |
| Mar. 2017 | A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness) | Southern Regional Honors Conference |
| Apr. 2016 | A Mathematical Model of African Sleeping Sickness | KSU Symposium of Student Scholars |